Detrital Magnetization in Red Beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona

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Specularite, apparently formed and in part altered to red hematite before deposition, carries much of the stable magnetization in sandstone, siltstone, and claystone of the Moenkopi Formation (Lower Triassic) at Gray Mountain, northern Arizona. High-temperature exsolution textures seen in polished section indicate that some, if not much, of the specularite is detrital. Progressive thermal and chemical demagnetization analyses indicate that the most stable magnetization resides in specularite. Last, a systematic depositional inclination 'error' is seen in both normally and reversely polarized sandstone. Part of its cause was the deposition along fine bedding laminations of subtabular grains of specularite carrying a strong direction of magnetization in the plane of tabularity. Magnetic inclinations are significantly shallower in sandstone than in siltstone, and the amount and sense of inclination error are controlled in detail by the dip of the horizontal and cross laminations. The foregoing leads to the conclusion that the stable magnetization is a detrital magnetization acquired during the course of deposition and that the Moenkopi Formation thus is suitable for magnetostratigraphy. Neither the formation of specularite nor the acquisition of the stable magnetization appears to be the consequence of prolonged and complicated postdepositional chemical alteration, a process invoked by some workers for formation and magnetization of ancient red beds. Because specularite is common in other ancient red bed sequences, their stable magnetizations also may be partly or largely detrital.

INTRODUCTION

Fold and conglomerate tests show that the stable magnetization was acquired early and rapidly in claystone and siltstone of the Lower Triassic Moenkopi Formation in the Gray Mountain area, northern Arizona [*Purucker and Elston*, 1977; *Purucker et al.*, 1978]. This report, dealing with the same locality, summarizes the results of a microscopic and correlated magnetic study of sandstone, siltstone, and claystone. Results of this study indicate that the stable magnetization resides mainly in grains of detrital hematite (specularite) and subordinately in an early hematite pigment and that the stable magnetization of the sediments was acquired during the course of deposition.

Three concepts concerning the magnetization of ancient red beds, stated by *Nagata* [1961, p. 213], have long been widely held. They are the following: (1) the hematite (and thus the magnetization) is dominantly in red pigment; (2) most hematite (pigment) is not an original constituent of the rock but has been produced in place by postdepositional chemical changes; and (3) hematite pigment is more stable magnetically than black specular hematite grains, which contribute little to the natural remanent magnetization (NRM) of the rock. If the foregoing concepts are accepted as valid, the magnetization of ancient red beds is entirely postdepositional, and the question that needs to be answered is the time of postdepositional magnetization. On the other hand, work involving chemical demagnetization of red beds by *Collinson* [1967], *Park* [1970], and *Roy and Park* [1972] and an investigation by *Collinson* [1974] of the relative importance of pigment and specularite in the remanent magnetism of red sandstone have indicated a more important role for specularite in the magnetization of red beds.

Evidence for the detrital origin of a significant part of the stable magnetization is developed along three lines of inquiry: (1) microscopic observations which reveal that at least some, if not most, of the specularite is detrital and that the distributions of unaltered and altered specularite grains, and of pigment, are not obviously related to one another, (2) relation of magnetic properties of hematite grains and pigment to magnetic characteristics of sandstone, siltstone, and claystone, which show that specularite carries much of the stable magnetization, and (3) demonstration of the existence of a systematic inclination 'error' in sandstone that is controlled in detail by the attitude of bedding laminations, and a correlation between the amount of inclination error and the grain size of the specularite and host rock.

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PETROGRAPHY AND PARAGENETIC RELATIONS

Red beds of the Moenkopi Formation are among a group of ancient red beds of Precambrian, Paleozoic, and Mesozoic age that are highly similar in depositional and lithologic character. Hematite in the Moenkopi is contained in two principal forms: (1) comparatively large (commonly >10 μm) black grains called specularite and (2) very fine grained red pigment (<1 μm). Textural and paragenetic relations between grains, and between grains and pigment, were most easily observed in very fine grained (0.064–0.125 mm) sandstone of the lower massive sandstone (for stratigraphy and nomenclature, see *Purucker et al.* [1978]). Some of the same relations also were observed in the finer grained rocks (in siltstone, <0.064 mm, and in silty claystone containing tiny, $\lesssim$2-μm specularite grains).

Microscopic study involved reconnaissance with the binocular microscope, followed by examination of polished thin sections and polished surfaces using combinations of reflected and transmitted light under the petrographic microscope. Reflected light study was carried out both in air and under oil immersion. For the most part, thin sections were cut from 2.5-cm-diameter cores that had been collected for paleomagnetic analysis. Because the cores were drilled parallel to the bedding, bedding laminations could be identified and readily traced in the thin sections under the microscope. About two dozen sections were examined, most from the lower massive sandstone. A few polished thin sections were prepared after the acid leaching of samples of sandstone to observe the effects of the acid on the pigment and specularite.

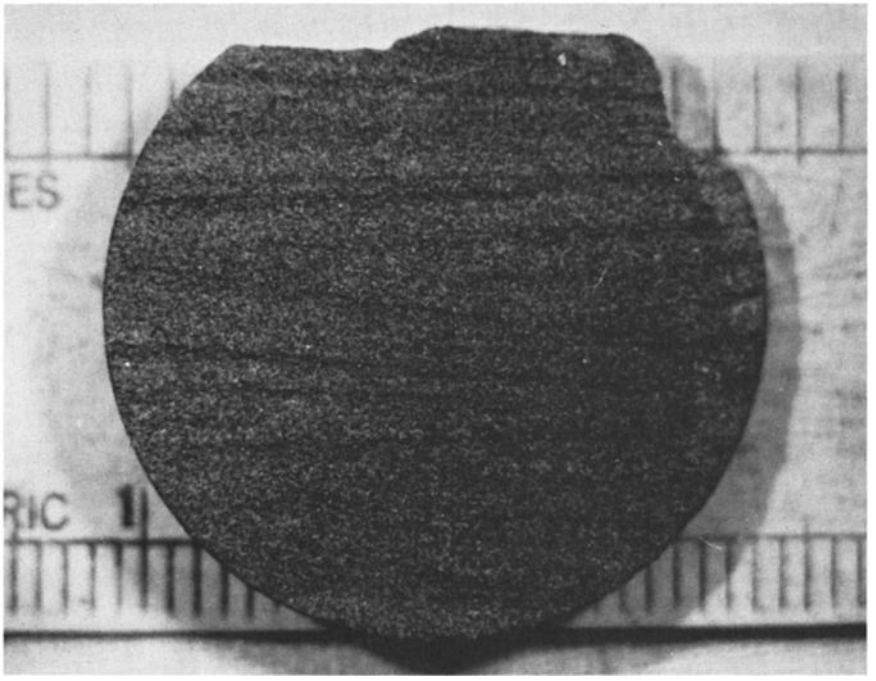


Fig. 1. Photograph of sample 1085 from upper unit of lower massive sandstone of the Wupatki member of the Moenkopi Formation showing fine cross-laminations enhanced by concentrations of black opaque mineral grains. Sample is from about 2 m stratigraphically above folded claystone bed of the middle unit of the lower massive sandstone [*Purucker et al.*, 1978, Figure 2]. Sample is 2.5 cm across.

The object of the study was to investigate (1) textural relations between iron-bearing grains and the fine-scale stratification and (2) paragenetic relations that might shed light on the time of formation and alteration of hematite grains and formation of hematite pigment. Paragenetic relations between grains, and grains and matrix, are physical and mineralogical interrelationships that provide evidence as to the order of formation and alteration of grains and minerals in a rock. Summarized below are the lithologic and mineralogic characteristics and the paragenetic relations seen principally in sandstone and subordinately in siltstone and claystone.

The lower massive sandstone is a well-sorted, silty, very fine grained sandstone. Angular to subangular and subrounded quartz is the dominant detrital mineral. Subordinate detrital grains are white mica, chlorite, carbonate, and opaque minerals. Specularite is the dominant opaque mineral, and zircon is common in detrital heavy mineral laminations that are rich in specularite. The sandstone is well bonded by silt-sized particles and sparse carbonate cement. Samples obtained from fairly fresh cliff faces exhibit only a thin altered zone (1–5 mm) related to present-day weathering. Behind this, fresh sandstone is mostly a rather even pale red (5R 6/2) to moderate red (5R 5/4), although some individual beds are a pale reddish brown (10R 5/4). The pale red color is imparted by a diffuse impregnation of hematite pigment. This comparative paucity in pigment (with respect to siltstone and claystone) has allowed paragenetic relations between pigment and specularite grains to be observed. Under the microscope the pigment is seen to be unevenly distributed in the sandstone, and variably about one third to one half of the silty intergranular matrix appears uncolored by pigment.

The sandstone is variously cross-bedded and cross-laminated and horizontally bedded and laminated. Cross-bedding appears to be of both eolian and fluviatile origin. Even in near horizontally stratified sandstone there are fine-scale cross laminations (see Figure 1). From studies of *Baldwin* [1973] and from our studies the lower massive sandstone is considered to be a beach deposit. Siltstone and sandstone beds that underlie the lower massive sandstone were deposited in generally quiet water, probably tidal mud flats as indicated by the presence of locally abundant mud cracks and salt casts.

Black to red opaque minerals make up about ½% or less of the lower massive sandstone. Many of the coarser opaque grains are concentrated in depositional laminae, some of which are several grains thick. These heavy mineral laminations range from horizontal and parallel laminated to arcuate and cross-laminated (Figure 1). Under moderate (×70) and higher magnifications (Figure 2), the opaque grains characteristically are seen to have crisp grain-to-grain contacts with adjacent black and red opaque grains and with enclosing quartz grains. No obvious alteration halos of red hematite pigment enclose the red grains or stain the silty matrix. The distribution of red pigment cannot be related in detail to the distribution of the reddish, partly to highly altered opaque grains.

The large opaque grains range mainly in size from 20 to 80 μm, although some are as large as 100–125 μm. These grains are in the size range of silt and very fine grained sand. Under moderate magnification a number of the larger opaque grains can be seen to be elongate or subtabular, and the long directions lie in or near the plane of the lamination (Figure 2*a*). This indicates that the tabular shapes controlled the depositional attitudes and that the grains were tabular at the time of deposition. Because specularite has a pseudocleavage that results in the production of subtabular grains during mechanical degradation of a crystal, the presence of such forms along bedding laminations suggests that the opaque grains were specularite when deposited.

Under the reflecting microscope at ×125, ×320, and ×800 magnification the great majority of opaque grains can be seen

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to be specular hematite, or specularite (Fe_2O_3), which displays pseudocleavage, high relief, and distinct anisotropy. The pseudocleavage results in a tabular to thin tabular aspect to many of the grains (Figure 2*a*). Rarely, a grain outline displays a partial octahedral form that reflects a pseudomorphous replacement of magnetite (martite). More commonly, a cross-hatched octahedral pattern, seen best under partly crossed nicols, reveals that the grain is polycrystalline and that magnetite (Fe_3O_4) was precursor to the specularite. Magnetite relicts in such specularite are rare (Figure 2*b*). Some specularite grains display no obvious structure but rather appear to be optically homogeneous. Other grains contain thin parallel light brown bands identified as ilmenite ($FeTiO_3$) exsolved in titanohematite (Figures 2*c* and 2*d*). Such textures are evidence for a high-temperature origin and indicate that these grains were hematite at the time of deposition. Additionally, some grains have widely differing reflectances, in some cases occurring within the same grain (Figure 2*e*), and the differences may reflect differing titanium contents. In a single heavy mineral lamination a wide variety of opaque and nonopaque grains are in random and sharp contact with one another (Figure 2), a situation expected for a heavy mineral lamination.

Individual specularite grains exhibit various degrees of alteration. Many grains show no apparent alteration or else the barest hint of a red patina on the black grain surface. Some specularite grains have narrow coatings or rinds of red pigment, and in such cases the pigment in the rinds commonly does not extend into the adjacent nonopaque matrix. A number of opaque grains exhibit a skeletal structure of specularite, within which are fillings of red hematite. In some cases the opaque grain is entirely, or nearly entirely, red hematite. Even these grains exhibit no red halos in the enclosing nonopaque matrix. Typical sharp contacts between fresh and altered grains are shown in Figure 2.

Some specularite is postdepositional, seen as overgrowths on unaltered and altered grains (Figures 2*d* and 2*e*). Some of the small pointed terminations are in optical continuity with specularite of the main grain, and some are not. These delicate

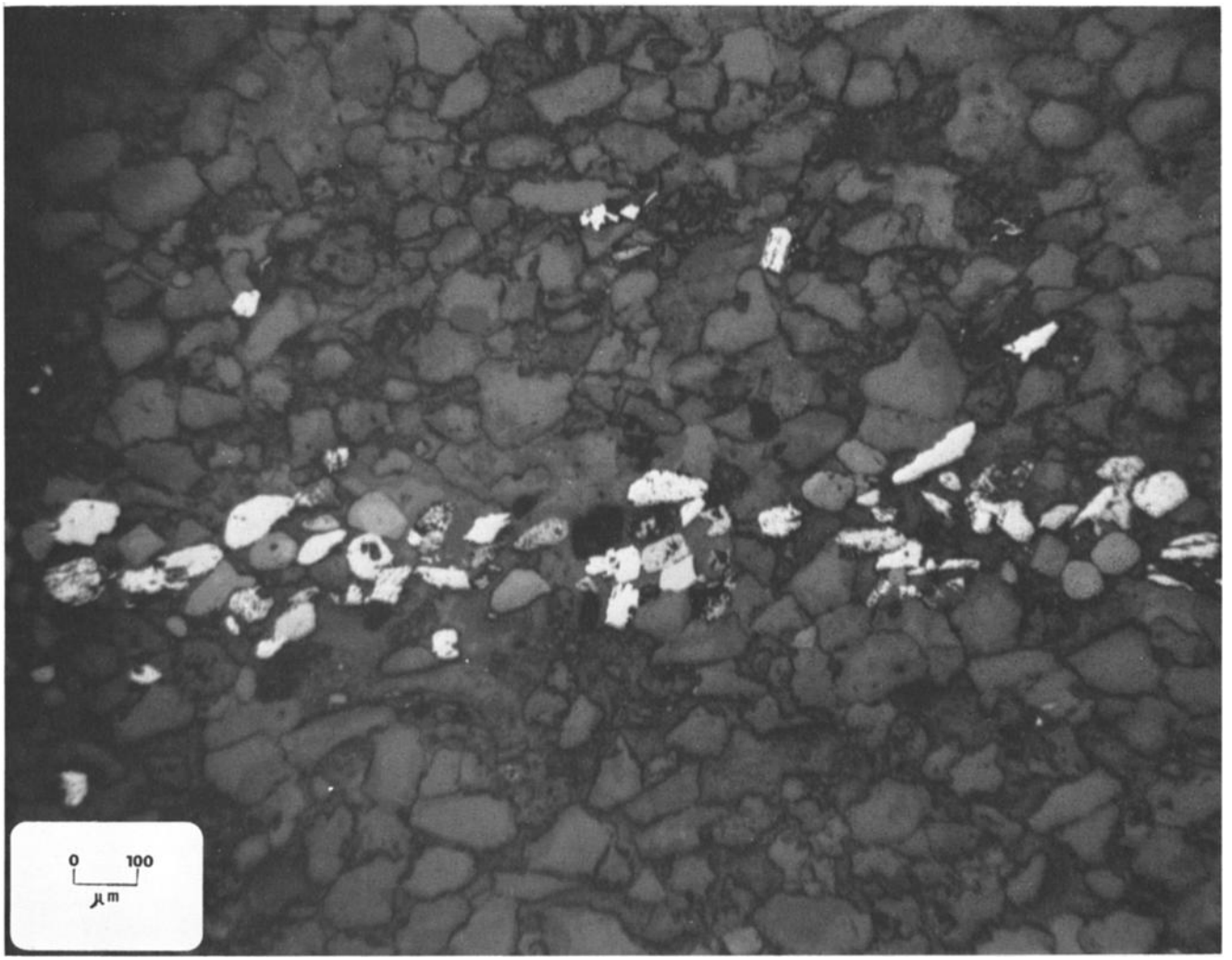


Fig. 2. (*a–e*) Photomicrographs of a polished thin section from horizontally laminated sandstone in upper unit of lower massive sandstone, Wupatki member of Moenkopi Formation (Triassic), Gray Mountain, Arizona. Photographs show distribution and contact relations of unaltered and altered grains of specularite in a single heavy mineral lamination. All photographs taken in plane reflected light. (*a*) Detrital lamination showing elongate and subtabular and angular to subrounded grains of specularite (white and very light gray) lying along plane of stratification. Grains partly to nearly completely altered to red hematite (mottled medium to very dark gray) appear randomly distributed along heavy mineral lamination and impart a bimodal aspect to the population of heavy minerals. Medium gray grains enclosing lamination are mainly quartz. Photograph taken in air.

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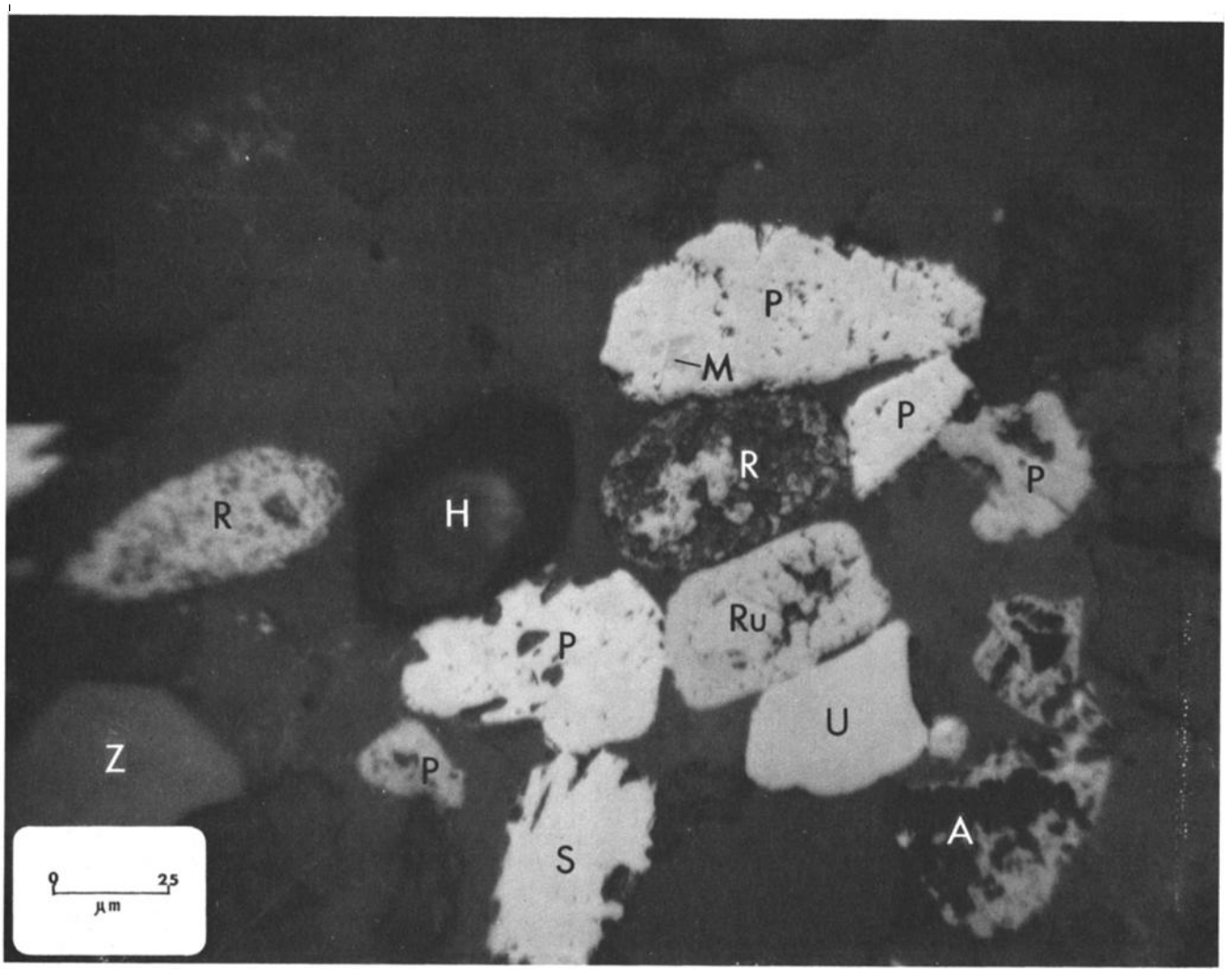


Fig. 2*b*. Enlargement of central part of Figure 2*a*, showing unaltered and little altered subangular to angular specularite grains (white and light gray) around and in sharp contact with rounded, highly altered red grain (dark and mottled light gray grain in right center of scene). P, polycrystalline specularite (martite) originally magnetite-ilmenite; polycrystalline grain in upper right contains relict magnetite (M). S, mainly single-crystal specularite. Ru, rutile or titanium-rich oxide. A, skeletal polycrystalline specularite altered in part to red hematite. R, red hematite grains, probably alteration products of specularite, Z, zircon. U, unidentified nonopaque grain. H, hole in section (plucked grain). Photograph taken in air.

features are evidence for the secondary precipitation of specularite, apparently after deposition. The effect of this specularite on the magnetization is not known, but because it is not volumetrically important, its contribution to the magnetization should be small. The existence of secondary specularite, however, is clear, whereas evidence for the postdepositional alteration of specularite to red hematite was looked for but not unambiguously recognized.

In addition to the silt-sized specularite grains, very tiny angular grains of specularite ($\lesssim 2$ μm) are irregularly disseminated in sandstone. Grains of similar size also are found in claystone and siltstone from within and beneath the lower massive sandstone. These tiny grains are most simply explained as fragments derived from larger grains of specularite fragmented during transport to the site of deposition. Although specularite is volumetrically small in siltstone and claystone, thermal analysis of siltstone and claystone and the acid leaching of sandstone, discussed later, reveal that specularite is an important carrier of the stable magnetization.

Magnetic Characteristics

The relative importance of pigment and specularite in the remanent magnetism of red sandstone and siltstone has been investigated by *Collinson* [1974]. He separated specularite and pigment from several samples of red beds of Permian and Triassic age, including a sample from the Moenkopi Formation. Grain size of the specularite employed presumably was in the range described in the preceding section for the Moenkopi Formation. After depositing the specularite and pigment in a magnetic field to produce an artificial detrital remanent magnetization (DRM), Collinson subjected the separate fractions and the parent rock to progressive thermal demagnetization. His demagnetization curves for a sample from the Moenkopi Formation are reproduced in Figure 3*a*, and they are similar in form to curves he obtained from several of the other red bed units he studied. As shown by Collinson, specularite is magnetically much 'harder' than pigment. The decay curve for whole rock lies close to the decay curve for specularite (Figure 3*a*), indicating that only a small proportion of the magnetization resides in pigment. This is explained because pigment very likely was not abundant in the sample of sandstone that he analyzed, similar to the lower massive sandstone which contains much less pigment than does associated siltstone and claystone.

Demagnetization curves for sandstone, siltstone, and claystone from Gray Mountain (Figure 3*b*) are shown for com-

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parison with the curves of Collinson. Although the Gray Mountain samples were collected from the freshest exposures, the samples nonetheless commonly show some effects of weathering. For sandstone, and particularly for siltstone and claystone, large decreases in intensity occur between the NRM and 400°C demagnetization steps. This decrease is accompanied by a shift to the Triassic normal polarity direction from an uncleaned NRM position that lies between the Triassic and the present-day field directions. Such shifts in directions are seen in siltstone and claystone as well as in sandstone. The shift reflects removal of much or nearly all of the secondary magnetization acquired by weathering in the present-day field. Oriented drill cores of the Moenkopi Formation obtained from well behind the surface of erosion have been shown to contain no present field component of secondary magnetization [*Baag and Helsley*, 1974]. In light of the lesser magnetic hardness of pigment the secondary components of magnetization presumably reside in late formed pigment, a pigment similar microscopically to early formed pigment.

At the 400°C and higher demagnetization steps (Figure 3*b*) a marked difference in intensity exists between the sandstone and the siltstone and claystone. The sandstone shows about a 40% decrease in intensity, whereas the claystone and siltstone show decreases of about 65–70%. The larger decrease in the fine grained rocks presumably reflects removal of magnetization that resides in the greater proportions of pigment. From 400°C to 600°C a gradual decrease of intensity is observed for all three materials. The gradual decrease appears characteristic of the decay of magnetization in specularite (Figure 3*a*) and is unlike the decay curve for the pigment. *Collinson* [1974, p. 263] concluded that specularite is the most important constituent in many red bed formations, contributing a very stable magnetization of depositional origin. Evidence from the Moenkopi Formation of Gray Mountain agrees with Collinson's experimental work and supports his conclusion.

CHEMICAL DEMAGNETIZATION

Three representative samples of sandstone from the lower massive sandstone in the Wupatki member were subjected to leaching in concentrated hydrochloric acid for the partial removal of hematite pigment. The object of such chemical demagnetization, as described by *Collinson* [1967], *Park* [1970], *Roy and Park* [1972], and *Reeve* [1975], is to observe changes in the direction and intensity of magnetization with the progressive removal of pigment. With complete removal of pigment the direction observed would be the direction of magnetization of the specularite grains, which resist dissolution by acid more than does pigment.



Fig. 2*c*. Compound detrital grain (center) having unaltered (upper left) and altered parts in sharp contact with each other and with adjacent, essentially unaltered specularite grains. In unaltered part of grain, thin brown (medium gray) parallel bands are probably exsolved ilmenite, which would be indicative of a high-temperature origin. In altered part of grain, red hematite occupies interstices of corroded grain, probably an iron-titanium oxide. Photograph taken in oil.

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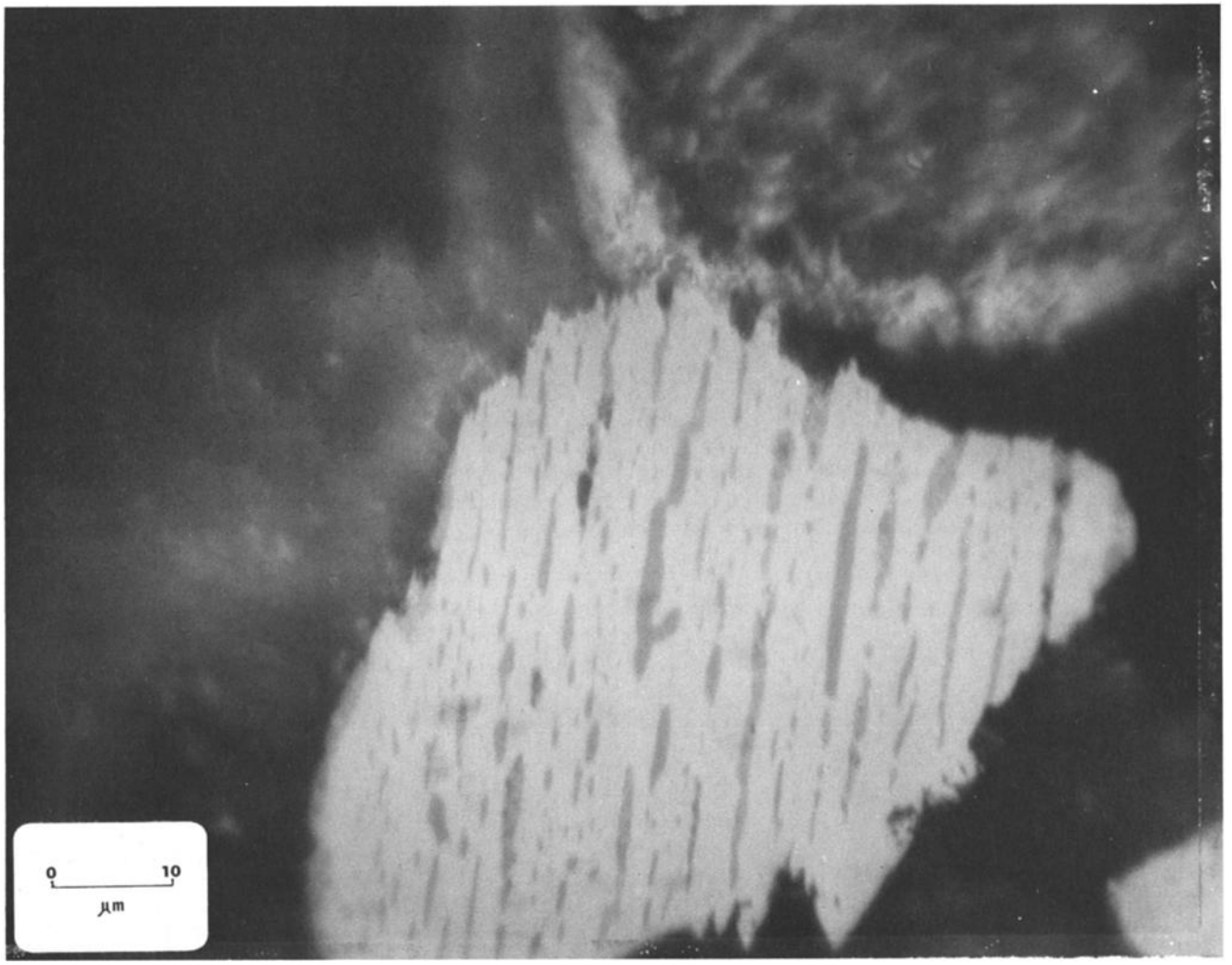


Fig. 2*d*. Specularite grain (light gray) containing bands of exsolved ilmenite (medium gray) indicative of high-temperature origin. Triangular projection on lower right and some of the other smaller triangular projections are secondary overgrowths of specularite. Altered grain in upper right, originally a titanomagnetite, has a rim of specularite. Photograph taken in oil.

The results of the partial removal of pigment by chemical leaching are shown in Figure 4. In 9–12 hours of leaching the intensity of magnetization decreased by nearly a factor of 2. Directions from two of the samples moved toward a Triassic normal polarity direction. Directions from the third sample (sample 1037) remained essentially unchanged. However, this sample came from the outer part of a core, very near the surface of weathering, and its anomalous direction and failure to respond to the chemical leaching may be due to the degree of recent weathering. Microscopic examination of the three samples under transmitted and reflected light revealed that appreciable amounts of the hematite pigment were removed by acid leaching and that the specularite appeared unaffected.

For sample 1085 (Figure 4) the removed pigment contained a component of magnetization that gave rise to an NRM direction which lay between the present field direction and the Triassic normal polarity direction. In the sample the removal of the present field component of magnetization by leaching paralleled its removal by thermal demagnetization. The path obtained from the chemical demagnetization of sample 1046, which for some reason responded better to chemical than to thermal demagnetization, approaches but does not reach a Triassic normal polarity direction. The results indicate that at least some secondary magnetization attributable to weathering resides in pigment that is removed by acid leaching.

INCLINATION AND BEDDING ERROR

The physical environment at the moment of deposition of magnetic particles in a sediment and the physical shapes of the particles (shape anisotropy) can cause deviations in their magnetic alignment with the ambient field. The former can involve the gravitational settling of particles in still water, the rolling of particles at the water-sediment interface, and the attitude of the surface of deposition on which the particles settle. Particle shape also can give rise to deviations if (1) some proportion of the magnetic particles are platy, (2) there is a relationship of platiness to anisotropy of magnetization, and (3) the platy grains, deposited subparallel to the bedding because of their platiness, are sufficiently large to be trapped in grain-to-grain contact, inhibiting postdepositional rotations. However, in sediments that remain water saturated for some time after deposition and in which the magnetic particles are considerably finer grained than the enclosing matrix, deviations from

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the ambient field can be mitigated by postdepositional rotations. That postdepositional rotations of magnetic grains (both magnetite and hematite) in a saturated sediment can occur in a matter of hours has been demonstrated experimentally by *Irving and Major* [1964]. *Irving* [1957], from study of red Precambrian Torridonian sandstone, has shown that the magnetization is carried in fine detrital particles and that slumped beds within the section have a uniform direction of magnetization, implying that the magnetic carriers were free to rotate in water-filled voids. Irving called this process postdepositional DRM.

Experiments and observations concerning depositional and postdepositional processes in the magnetization of sediments have been summarized in a recent review by *Verosub* [1977]. Deposition of natural material from suspension of varved clay revealed that declination was consistent with the applied field but that inclination was as much as 20° too shallow [*Johnson et al.*, 1948]. A similar experiment with powdered Triassic red sandstone (Lloyd, quoted in the paper by *Clegg et al.* [1954]) produced an 8° deviation, an effect that *King* [1955] called 'inclination error.' King found an empirical relationship between inclination of the applied field and inclination of the sediment, and he explained the inclination error by assuming that the sediment contained a fraction of disc-shaped particles whose magnetic moments resided in the plane of the discs. *Griffiths et al.* [1960] subsequently extended the theory to include spherical particles.

Laboratory experiments of *King* [1955] also revealed that the direction of magnetization of sediments is influenced by the slope (or dip) of the bedding plane, an effect he named 'bedding error.' Bedding error is defined as the difference between the magnetic inclination observed on a dipping bed and that observed on a horizontal bed. It later was found to be approximately equal to the slope of the bedding plane [*Hamilton and King*, 1964].

Last, King [1955] recognized that deposition in flowing water affected the direction of magnetization in a sediment, and *Granar* [1958] showed that if the flow is oblique to the field, errors in both inclination and declination will result. Quantitative measurements of the 'current rotation' effect were made by *Griffiths et al.* [1960] and *Rees* [1964].

Our studies in the Moenkopi Formation have shown that (1) a substantial inclination error exists between sandstone and siltstone, (2) there is a systematic and predictable bedding error in cross-laminated sandstone, documented in both normally and reversely polarized rock, and (3) deviations in decli-

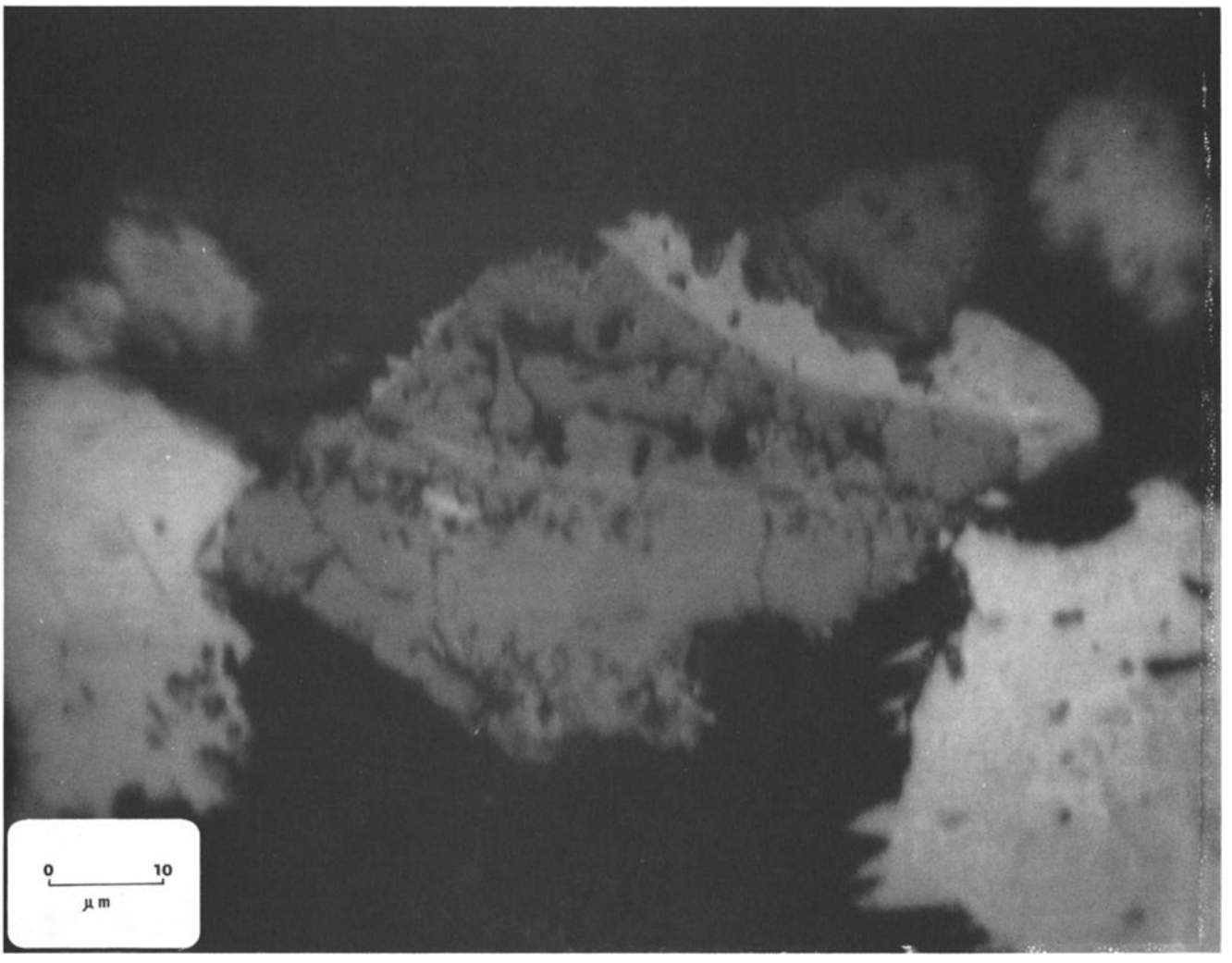


Fig. 2*e*. Grain of altered titanium-rich oxide (medium gray) containing band and seam of optically homogeneous specularite (white) and small triangular overgrowths of specularite. Polycrystalline specularite grains are to left and right. Photograph taken in oil.

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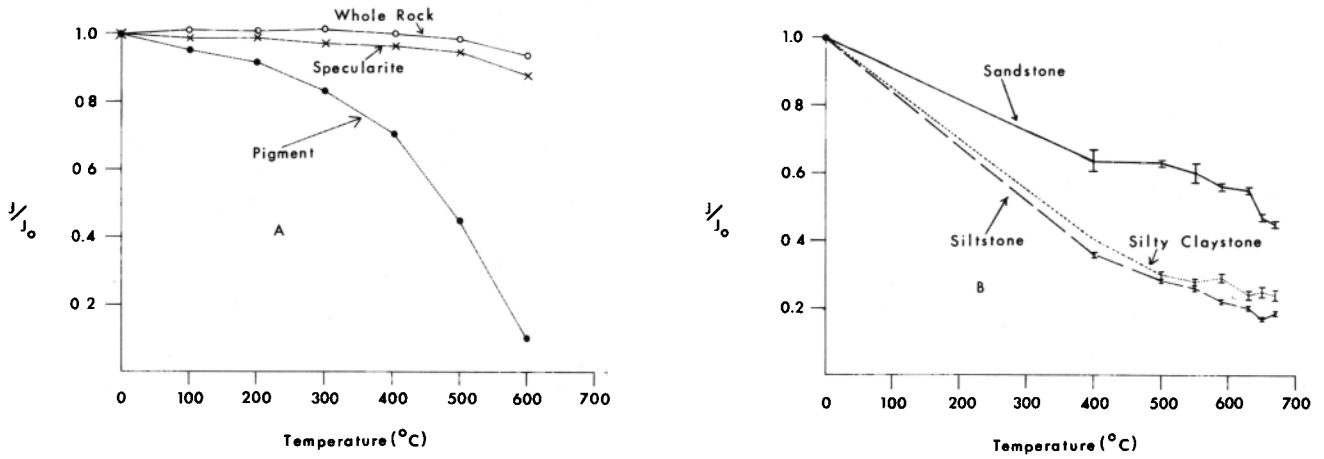


Fig. 3. Plot of intensity normalized to the natural remanent magnetization (NRM) for samples subjected to progressive thermal demagnetization. Figure 3*a* shows sandstone of the Moenkopi Formation analyzed by *Collinson* [1974, Figure 2]. Figure 3*b* shows demagnetization curves for sandstone of lower massive sandstone (N = 46), siltstone clasts in base of lower massive sandstone (N = 15), and folded silty claystone from within the lower massive sandstone of the Wupatki member (N = 29). Error bars represent standard deviations about the means.

nation are seen in near horizontally laminated sandstone and that the deviations appear to be attributable to current flow.

Magnetic Inclination in Sandstone and Siltstone

The lower massive sandstone, 10 m thick, and an underlying section that consists principally of siltstone, 26 m thick, have been sampled extensively on close (2–70 cm) stratigraphic spacings and in multiple overlapping sections. Results, summarized in Table 1, reveal that magnetic directions in sandstone and siltstone differ at greater than 99% confidence level and that the difference is almost entirely in inclination. The inclination is 9°–15° shallower in sandstone than in siltstone. The magnetic direction in the siltstone agrees closely with the mean direction reported for the Moenkopi Formation of Colorado and Utah [*Helsley and Steiner*, 1974]. As can be seen in Table 1, substantial differences in inclination exist between sites in the sandstone, suggesting that some property of the sandstone unrelated to stratigraphic position has affected the magnetic inclination. Secular variation has been looked for, but not recognized, in the stratigraphically controlled samples from the siltstone and sandstone. In light of bedding inclination tests, discussed in the following section, we conclude that secular variation was not important and cannot be appealed to for explanation of the shallow inclinations seen in sandstone.

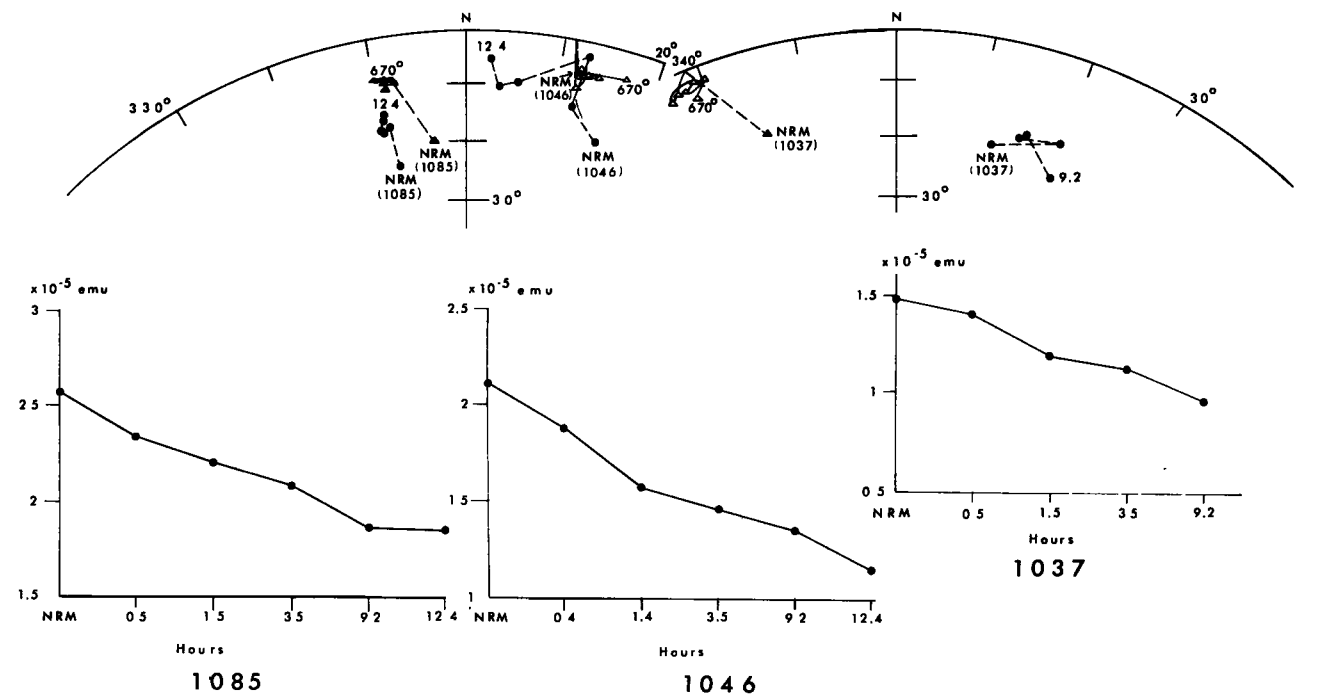


Fig. 4. Directions and intensities for three samples subjected to chemical leaching (circles) plotted against time, and directions for corresponding samples obtained during progressive thermal demagnetization (triangles). Sample numbers are shown beneath the individual plots. The location of sample 1046 is shown by *Purucker et al.* [1978, Figure 5]. On equal-area plots, solid symbols are the lower hemisphere, open symbols are the upper hemisphere, emu is electromagnetic units.

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TABLE 1. Mean Directions of Magnetization Derived From Sandstone and Associated Siltstone and Claystone of the Moenkopi Formation, Gray Mountain Area, Arizona

	$\bar{D}$, deg	$\bar{I}$, deg	k	R	α_{95}, deg	α_{99}, deg	N
			Sandstone				
3 sites, 550°C	345.7	8.2	27	126.1323	2.4	3.0	131
1 site, 590°C	345.7	3.4	27	43.4003	4.1	5.2	45
			Siltstone (With Minor Claystone)				
550°C	343.5	17.1	30	82.2410	2.8	3.5	85

Directions for the sandstone and siltstone are statistically different at 99% confidence level using the f ratio statistic of *Watson* [1956], described by *Irving* [1964, p. 62]. $\bar{D}$ is mean declination; $\bar{I}$ is mean inclination; k is *Fisher*'s [1953] precision parameter $(N - 1)/(N - R)$, where R is the resultant of N directions of magnetization, expressing the scatter of the data points on a spherical surface; α_{95} is radius of cone around the mean direction drawn at the 95% confidence level; N is the number of samples.

Bedding Inclination Error

The anomalously shallow inclinations seen in the lower massive sandstone and the presence of subtabular heavy mineral grains lying along bedding laminations in the sandstone have given rise to a field test for bedding inclination error. If the opaque grains were specularite when deposited, suggested by subtabular shapes that parallel the bedding, then they would contain a strong direction of magnetization in the direction of the basal plane which, in turn, could give rise to a systematic bedding inclination error. Additionally, the rolling of spherical particles at the sediment-water interface can contribute to the bedding inclination error [*Griffiths et al.*, 1960], although the effect would be less pronounced. If the inclination error can be related in a predictable and detailed way to the dip of sedimentary laminations, then the magnetization would be of detrital origin, acquired as each lamination was deposited in an aggrading body of sand. If the magnetization is detrital, the test should prove positive for both normally and reversely polarized sandstone.

In the normally polarized lower massive sandstone, three closely adjacent sites were selected that contain near horizontally laminated sandstone (H), southerly dipping cross-laminations (S), and northerly dipping cross beds (N). The detailed character of the horizontal laminations can be seen in Figure 1. Site H directly overlies site S and is within a single bed of sandstone. Site N is essentially at the same stratigraphic horizon but a few meters to the north. The character of the stratification, stratigraphic relations, and the distribution of samples are shown in Figure 5, as are equal-area plots of the NRM and cleaned directions for the individual samples. The mean directions, with statistics, are summarized in Table 2. NRM directions for the group N samples lie near the present field direction. Samples from groups H and S came from a fairly fresh face formed by spalling, whereas group N samples came from a partly eroded ledge in which the red color of the rock appears more diffuse and less fresh. The small shifts away from the present field direction of group H and S samples with cleaning and the large shift for group N samples support the field observations.

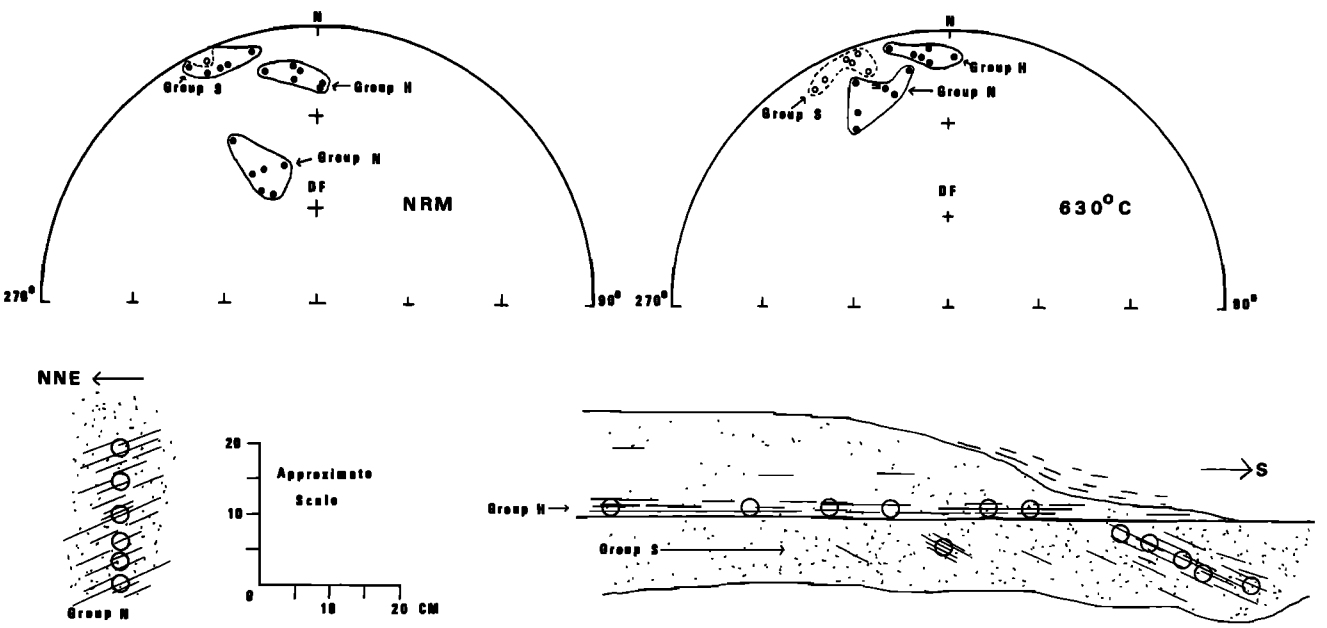


Fig. 5. Distribution of samples in horizontally laminated and cross-laminated sandstone and paleomagnetic directions from these normally polarized samples. Paleomagnetic convention assigns negative inclinations to upward pointing magnetic vectors and positive inclinations to downward pointing magnetic vectors. Equals sign is mean normal polarity direction for the Moenkopi Formation [*Helsley and Steiner*, 1974]. DF is geocentric dipole field. Locations of sample groups are shown by *Purucker et al.* [1978, Figure 2]. Means for sample groups are listed in Table 2.

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TABLE 2. Mean NRM and Stable Directions of Magnetization for Normally Polarized Samples Collected From Horizontally Laminated and Cross-Laminated Strata in Upper Unit of Lower Massive Sandstone of the Wupatki Member of the Moenkopi Formation, Cedar Wash, Gray Mountain Area, Arizona

	$\bar{D}$, deg	$\bar{I}$, deg	k	R	α_{95}, deg	N
			Group H (Horizontal)			
NRM	355.4	18.7	185	5.9729	4.9	6
550°C	354.5	8.7	276	5.9819	4.0	6
630°C	353.9	9.2	230	5.9783	4.4	6
			Group S (Strike 86°–113°; Dip 19°–26°S)			
NRM	337.2	5.5	126	5.9603	6.0	6
550°	336.4	−6.5	187	5.9733	4.9	6
630°C	336.4	−6.4	191	5.9737	4.9	6
			Group N (Strike 300°; Dip 20°–24°N)			
NRM	337.1	47.4	102	5.9511	6.7	6
550°C	339.6	21.4	84	5.9406	7.3	6
630°C	340.2	20.1	91	5.9448	7.1	6

Data points are plotted in Figure 5. See Table 1 for explanation of symbols.

A test for bedding inclination error has also been carried out on reversely polarized sandstone (Figure 6 and Table 3). Less than a meter below the lower massive sandstone is an interval of reversed polarity that is slightly more than 1 m thick [*Shoemaker and Purucker,* 1974]. It includes thin interbeds of siltstone and silty sandstone. Beds of very fine grained, finely laminated silty sandstone locally occupy much of the interval of reversed polarity. One of these beds, exposed in a northwest-southeast cliff face, was sampled for the test. The generalized character of the bedding, the dip of the laminations, and the distribution of samples are diagramed in Figure 6. The sandstone is finer grained and more silty than the overlying lower massive sandstone, and the entire unit is finely laminated. The laminations allowed the strike and dip to be determined for each sample. Most laminations dip moderately to the northwest and west. However, within the bed of sandstone, small areas can be found in which the attitudes of the laminations reverse abruptly and dip south. The largest of such areas are of the order of a decimeter in lateral extent and a few centimeters to a decimeter in height. Some are hardly larger than the 2.5-cm diameter of a core sample. The areas of dip reversal are a measure of the local relief on small climbing ripples that existed during the aggradation of silty sand that forms this bed.

The test for bedding inclination error for both normally and reversely polarized sandstone proved positive. Each of the groups of samples exhibit distinct, nonoverlapping directions. Magnetic inclinations behave as predicted if it is assumed that bedding plane control of magnetization exists.

A systematic relationship exists between bedding error β and dip of the bedding surface, α (Figure 7). A least squares fit to a straight line that passes through the origin yields a ratio $\beta/\alpha = 0.76$. This value is close to the ratio that can be calculated from theoretical considerations [*Hamilton and King,* 1964]. Using a Triassic inclination of 18° [*Helsley and Steiner,* 1974] and an inclination error of 11° (9°–15° inclination error is seen in horizontally laminated sandstone, Table 1), then the theoretical value $\beta/\alpha \cong 0.8$ [*Hamilton and King,* 1964, Figure 3].

Last, laminations that dip steeply toward and away from the direction of the Triassic field have declinations that lie close to the mean Triassic declination. However, laminations that have low dips (group H, Figure 5; group N, Figure 6) exhibit declinations that are shifted. Northwest to southeast trending current lineations are present on parallel laminated sandstone at the horizon of group H samples in the lower massive sandstone (Figure 5). Currents presumably were responsible for the shift in declination in the group H samples. The shift in declination in the group N samples in the reversely polarized sandstone (Figure 6) presumably also reflects current flow.

DISCUSSION

Microscopic observations and the character of magnetization of specularite together provide an explanation for the existence of a bedding inclination error in sandstone. A number of the larger specularite grains tend to be elongate and subtabular and lie subparallel to the bedding (Figure 2*a*). The larger grains are closely packed in grain-to-grain contact with grains of similar and larger size, and this packing inhibited postdepositional rotations. The magnetization of specularite is anisotropic, the strong (easy) direction of magnetization occurring along the direction of the basal plane. This is a direction that parallels the pseudocleavage [*Howell et al.,* 1960; *Fuller,* 1963; *Stacey and Banerjee,* 1974, p. 89] and thus presumably the subtabularity. Such an interrelationship of physical and magnetic properties appears in large part to be responsible for the observed bedding inclination error in the normally and reversely polarized sandstone.

Although specularite is a carrier of stable magnetization in both siltstone and claystone (Figure 3), bedding inclination error may not be important. In siltstone, specularite tends to be much finer grained than much of the host material, which would allow rotations in intergranular voids. Moreover, tabularity is not obvious. Opaque grains in silty claystone, which appear to be mainly specularite, are extremely fine ($\lesssim 2$ μm) angular fragments. Because they 'float' in a matrix of clay there should have been no impediment to their alignment with the ambient field before and during dewatering of the sediment. Postdepositional rotations in saturated sediment may well have taken place. Siltstone, deposited in a tidal flat environment, presumably remained saturated long after deposition of the individual beds. This contrasts to deposition on a beach (lower massive sandstone), where water could drain rapidly following deposition of individual laminations.

The existence of an inclination error in ancient specularite-bearing sandstone implies that wherever possible, poles should be calculated from finer grained siltstone and claystone. However, the presence of an inclination error arising from detrital specularite is not a feature to be entirely avoided during any paleomagnetic study of ancient red beds. Polarity is faithfully recorded. Moreover, there now is a field test which, if positive, shows that the stable magnetization is of detrital and depositional origin. The importance of this for establishing the validity of a pole and polarity zonation is obvious.

The Moenkopi Formation at Gray Mountain has been reported, in abstracts, to be unusable for magnetostratigraphy on both paleomagnetic and mineralogic grounds [*Larson et al.,* 1976; *Walker and Larson,* 1976]. Our results indicate just the opposite. The model for red bed formation that underlies the work of Walker and of Larson is found in the work of *Walker* [1967, 1974, 1976]. Because red beds analogous to the Moenkopi are not seen in present-day and upper Cenozoic desert deposits, and because a greater reddness can be observed in some Miocene strata, *Walker* [1976, pp. 279–281] has proposed that the reddening of ancient red beds occurred gradu-

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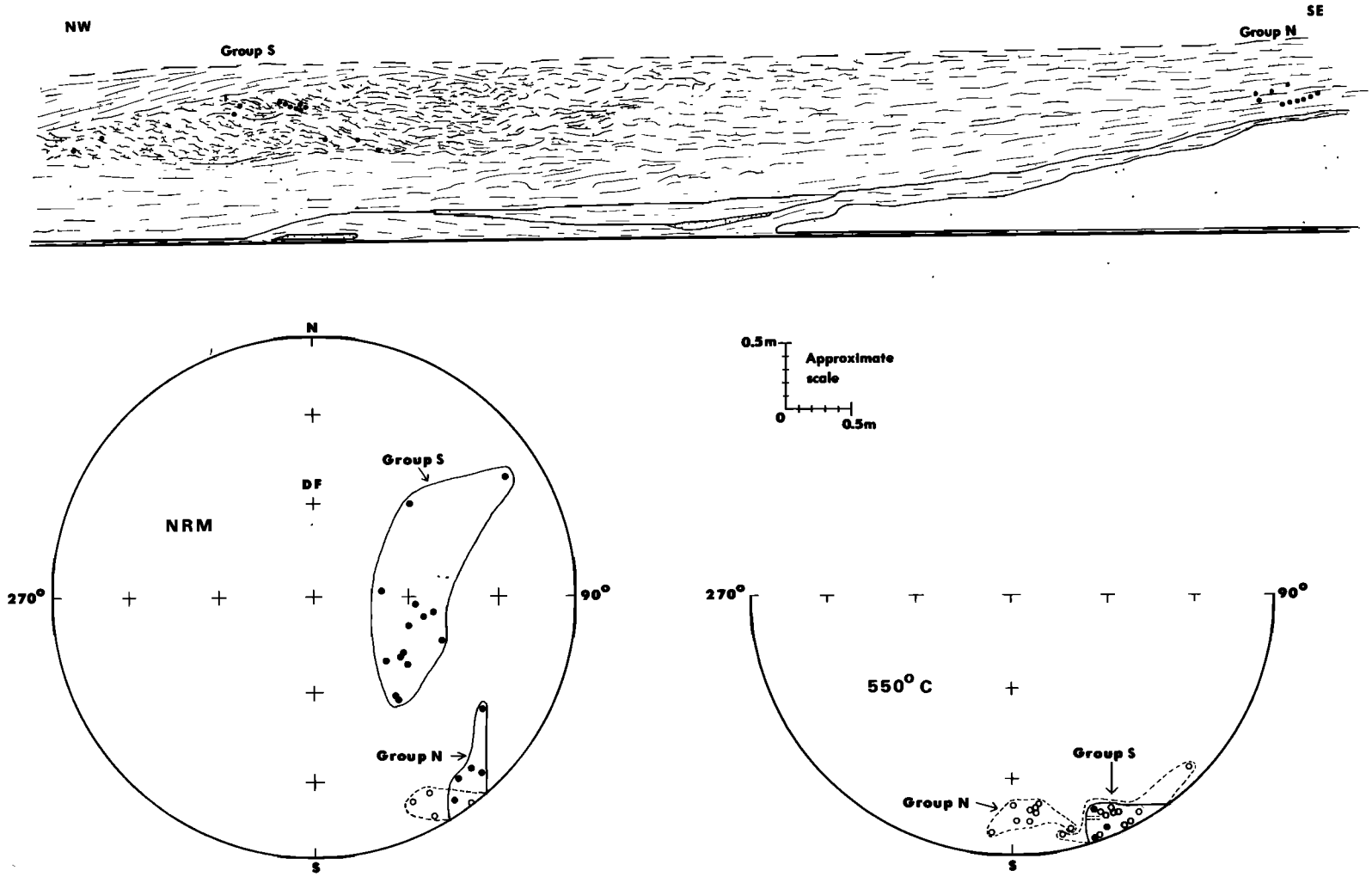


Fig. 6. Distribution of samples in finely laminated silty sandstone and paleomagnetic directions from these reversely polarized samples. Equals sign is mean reversed polarity direction for the Moenkopi Formation [*Helsely and Steiner*, 1974]. Locations of sample groups are shown by *Purucker et al.* [1978, Figure 3]. Means for sample groups are listed in Table 3. See Figure 5 for other abbreviations and conventions.

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TABLE 3. Mean NRM and Stable Directions of Magnetization for Reversely Polarized Samples Collected From a Laminated Sandstone Bed Within a Narrow Interval of Reversed Polarity That Closely Underlies the Lower Massive Sandstone, Cedar Wash, Gray Mountain, Arizona

	$\bar{D}$, deg	$\bar{I}$, deg	k	R	α_{95}, deg	N
		Group S (Mean Strike 91°; Dip 19°S)				
NRM	104.8	55.9	14	13.0568	11.0	14
500°C	155.9	−1.6	30	13.5677	7.4	14
550°C	156.1	−3.8	62	13.7908	5.1	14
590°C	155.4	−2.9	70	13.8141	4.8	14
630°C	160.7	−6.0	60	13.7844	5.2	14
		Group N (Mean Strike 175°; Dip 19°W)				
NRM	143.1	3.2	28	8.7106	10.0	9
500°C	173.0	−15.6	87	8.9084	5.5	9
550°C	175.7	−15.6	136	8.9413	4.4	9
590°C	175.9	−18.4	64	8.8743	6.5	9
630°C	181.8	−16.7	74	8.8919	6.0	9

Data points are plotted in Figure 6. See Table 1 for explanation of symbols.

ally over millions of years and that the coarsely crystalline hematite (specularite) seen in ancient red beds is the end product of prolonged postdepositional alteration. Such a model fully applies the principle of uniformitarianism and assumes that the present-day environment can be extrapolated unchanged to Triassic time.

Larson and Walker [1975] investigated the magnetization of upper Cenozoic desert deposits of Baja California. They found evidence for the postdepositional reddening of magnetite-bearing sediments and for the development of secondary magnetizations associated with hematite and other pigmenting material. From this the magnetization of ancient red beds is presumed by them to reside in red pigment and in specularite formed long after deposition.

We have found in the Moenkopi Formation evidence that specularite was formed and partly altered before it was incorporated in the sediments and that the environment of deposition appeared to be conducive to the precipitation of hematite pigment during the course of deposition. Such a situation implies that highly oxidizing conditions were prevalent both in the source terrane and in the depositional environment. A further implication is that the ancient environment was not like the present-day environment because analogous red beds are not being formed and deposited today. This then suggests that the earth's environment has changed through time and that a simple extrapolation of the present environment to the ancient past may not be valid.

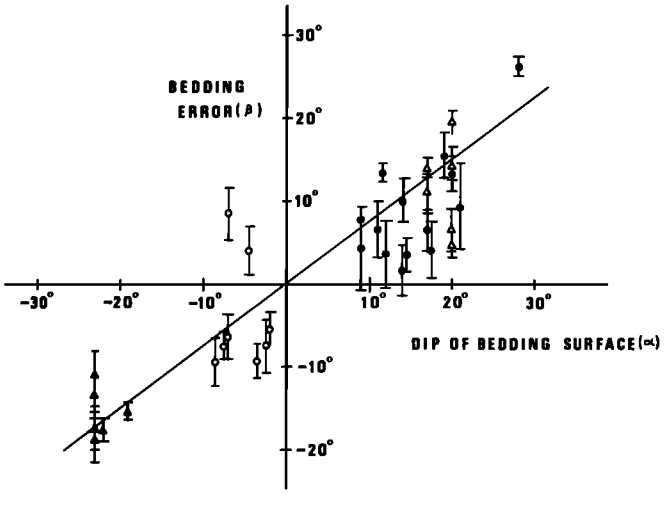


Fig. 7. Bedding error β versus dip of bedding surface, α, for normally (triangles) and reversely (circles) polarized sandstone. Open symbols represent north dipping laminations; solid symbols represent south dipping laminations. Bedding error is defined as the difference between the magnetic inclination observed on a dipping surface and that observed on a horizontal surface. The dip of the bedding plane is corrected for cases in which the Triassic magnetic meridian [*Helsley and Steiner*, 1974] is oblique to the direction of maximum slope. Bedding errors and associated error bars were derived, from the mean and standard deviation, respectively, of thermally cleaned directions (550°, 590°, and 630°C) listed in Tables 2 and 3. A least squares fit to a straight line that passes through the origin yields a ratio $\beta/\alpha = 0.76$.

SUMMARY

Our observations and experiments indicate that much of the stable magnetization in red beds of the Lower Triassic Wupatki member of the Moenkopi Formation at Gray Mountain, Arizona, resides in detrital specularite. The detrital magnetization is supplemented by an apparent early chemical magnetization that resides in hematite pigment. Holocene and Triassic components of secondary magnetization residing in pigment commonly can be removed at temperatures of 500°–590°C. This is supported by chemical leaching, which indicates that much of the secondary magnetization resides in late formed pigment.

Exsolution textures show that at least some of the specularite in the lower massive sandstone of the Moenkopi Formation is of high-temperature origin and is detrital. Some grains appear to have undergone partial to largely complete alteration to red hematite before deposition. Hematite pigment is diffusely and irregularly disseminated in the sandstone. It displays no systematic spatial relation to altered and unaltered specularite grains, and the altered and unaltered grains, in turn, appear randomly distributed along heavy mineral laminations. Magnetite relict in specularite is present but is not important magnetically, an observation supported by progressive thermal demagnetization analysis. The foregoing suggest rather thorough alteration of magnetite-bearing rocks in the source terrane. The environment responsible for the formation and alteration of specularite in the source terrane presumably also was responsible for formation of an early disseminated hematite pigment at the site of deposition.

Elongate to subtabular, silt-sized, and very fine grains of specularite are present along heavy mineral laminations. The tabular grains lie in the plane of the depositional laminations,

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where grain-to-grain contacts have inhibited postdepositional rotations. The tabular character suggests that the grains were specularite when deposited. The tabularity apparently is the result of pseudocleavage parallel to the basal plane of specularite, which also is the direction of strong (easy) magnetization. In very fine grained sandstone in the Moenkopi this interrelationship has resulted in magnetic inclinations controlled by the dip of bedding laminations. A systematic, predictable bedding plane control of magnetic inclinations is seen in both normally and reversely polarized sandstone. Magnetic inclinations are 9°–15° shallower in the very fine grained lower massive sandstone than in siltstone. Because an appreciable inclination error exists in sandstone, paleomagnetic poles should be calculated from directions obtained from finer grained rocks.

The foregoing argues for the predepositional formation and alteration of specularite and for the formation of early hematite pigment during the course of deposition of beds of the Triassic Moenkopi Formation. This implies strongly oxidizing conditions both in the source terrane and in the area of deposition. The Moenkopi is typical of ancient red beds, and many, if not most, Phanerozoic and Precambrian red beds that we have examined are specularite bearing. Because strata typical of ancient red beds seem to disappear from the geologic record by about the end of Jurassic time and because no strata typical of ancient red beds are being deposited today (detrital magnetite is the carrier of magnetization in most Cenozoic sediments), ancient red beds may reflect an environment that differed from today's environment in some way that cannot be deduced directly from the geologic record. Our earth is a continuously evolving body, and unless there is evidence for the extrapolation, simple extrapolation of the present environment to the distant geologic past may not be valid.

Acknowledgments. Initial sampling in 1971 of the section of Moenkopi Formation at Gray Mountain, Arizona, which led to identification of the Gray Mountain interval of reversed polarity, was carried out jointly with E. M. Shoemaker, supported by the Elston and Shoemaker families. We wish to thank S. Elston for assistance in the collection of additional samples and for maintaining the typescript through the various stages of preparation. S. Gillett assisted in some of the later sampling and also wrote some valuable computer programs. Reviews of the manuscript by R. Reynolds, C. S. Grommé, and J. Hillhouse of the U.S. Geological Survey and by reviewers for JGR are gratefully acknowledged. Additionally, we thank R. Brett, and especially R. Reynolds, for most helpful discussions on the microscopy of opaque minerals.

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